

NIIS HACKATHON 2026 — OFFICIAL PROBLEM BRIEF

Track: Neurocore / GeoNavX (Location Intelligence & Robotics)

Challenge ID: PS-12

Title: Emergency Route & Location Intelligence System

1. Executive Summary & Problem Overview

During severe urban crises, catastrophic climate disasters (such as urban flash floods, structural collapses, cyclone landfalls, and chemical gas leaks), and acute localized emergencies (such as urban fires or public safety threats), conventional commercial GPS navigation engines prove dangerously inadequate. Commercial mapping tools only optimize for typical passenger travel time; they do not possess real-time awareness of expanding disaster perimeters, submerged roads, fallen electrical wires, structural debris, or safe emergency shelter capacities.

Victims, evacuating families, and emergency response teams often find themselves trapped in gridlocks or routed directly through high-hazard disaster zones. Furthermore, municipal disaster management authorities lack unified spatial intelligence to enforce localized evacuation corridors, publish dynamic hazard geofences, and coordinate multi-agency resource logistics in real time.

The objective of **PS-12 (Emergency Route & Location Intelligence System)** is to architect an intelligent, location-aware emergency response and spatial evacuation platform. The system must ingest user location coordinates, calculate dynamic hazard-evasive safe evacuation routes, map nearby emergency facilities (trauma centers, blood banks, relief shelters), enforce automated risk geofences, and broadcast real-time, context-aware navigational guidance.

2. Core Personas & Access Roles

1. **Evacuating Citizen / Incident Reporter:** Triggers one-tap geolocated SOS alerts, receives real-time safe evacuation turn-by-turn guidance avoiding active hazards, and locates functional emergency relief shelters.
2. **Emergency Response Team (Ambulance / Fire / Disaster Rescue):** Accesses prioritized emergency corridors, vehicle-constrained navigation, and live updates on reported road obstructions.
3. **Disaster Management Incident Commander:** Creates and manages spatial danger zones (geofences), monitors evacuation density heatmaps, allocates emergency relief supplies, and broadcasts geo-targeted civil alerts.

4. **Relief Camp & Facility Coordinator:** Updates live facility availability metrics (beds, medical supplies, clean drinking water, food rations) to prevent overcrowded shelter allocations.

3. Key Functional Modules & Technical Deliverables

Module 1: Location Intelligence & Incident Ingestion

- **High-Precision Geolocation & SOS Dispatch:** Ingests user GPS coordinates, reverse-geocoded landmark references, and multi-tiered distress tags (Medical Trauma, Trapped / Submerged, Fire Hazard, Structural Collapse).
- **Offline-Resilient Emergency Beacon:** Caches local device coordinates and generates compact SMS or QR-based location payload strings when cellular internet degrades.
- **Crowdsourced Hazard Reporting:** Enables citizens and responders to report localized ground barriers (waterlogged roads, fallen trees, live power lines, debris) with automated spatial clustering.

Module 2: Dynamic Hazard Geofencing & Alert Engine

- **Polygon & Radial Risk Geofencing:** Empowers emergency commanders to draw dynamic polygonal risk zones (e.g., Flood Surge Perimeter, Fire Spread Boundary, Gas Leak Toxicity Radius) with automated expiration timers.
- **Proximity Violation Alerts:** Automatically detects if an evacuating user or responder is heading toward an active danger zone and fires high-priority audible and visual reroute warnings.
- **Location-Based Civil Defense Broadcasts:** Pushes geo-targeted emergency advisories to all registered devices located within a defined geographic boundary.

Module 3: Safe-Route Identification & Dynamic Rerouting Engine

- **Hazard-Evasive Route Planning:** Computes escape routes that actively penalize or strictly block traversal through active hazard geofences and reported road closures.
- **Multi-Criteria Evacuation Routing:** Balances distance against hazard clearance safety margins, avoiding low-elevation flood plains and choke points.
- **Dynamic Real-Time Rerouting:** Automatically re-evaluates and alters active escape navigation paths if an expanding hazard intersects the current traversal line.

Module 4: Emergency Facility Discovery & Live Logistics

- **Spatial Facility Mapping:** Displays interactive geospatial layers of verified hospitals, trauma centers, police stations, fire hydrants, and disaster relief shelters.
- **Nearest Functional Resource Allocation:** Algorithms that route evacuees not merely to the closest physical shelter, but to the optimal shelter factoring in live bed availability and flood-free transit accessibility.

- **Offline Vector Map Tile Fallback:** Client-side vector tile caching (IndexedDB / PouchDB) allowing users to view maps and navigate without active cloud server connectivity.

4. Recommended Technology Stack

Layer, Suggested Technologies

Frontend Web & Command Map, React.js (Vite) / Next.js / Tailwind CSS / Leaflet.js / Mapbox GL JS / Deck.gl / Lucide Icons

Mobile Application (PWA / App), "React Native, Flutter, or Responsive Progressive Web App (PWA) with Geolocation API"

Backend Architecture, Python (FastAPI / Flask) or Node.js (Express) with REST / WebSockets for live telemetry

Geospatial Database, "PostgreSQL with PostGIS / Supabase (Spatial geometries, ST_DWithin, ST_Contains, pgRouting)"

Routing & Spatial Algorithms, "OSRM / GraphHopper / Valhalla, Turf.js for spatial topology and polygon intersections, NetworkX"

Real-Time Data & Caching, "Redis Pub/Sub, WebSockets for live responder tracking and instant geofence sync"

5. Evaluation & Judging Criteria (100 Points Total)

1. **Safe-Route & Hazard Avoidance Precision (25 Points):** Demonstrable routing algorithms that successfully identify and reroute around active hazard zones, road closures, and danger perimeters.
2. **Geofencing & Spatial Intelligence Engine (25 Points):** Efficacy and responsiveness of the dynamic geofencing system, proximity warnings, and multi-agency facility mapping.
3. **Emergency UX, Accessibility & Responsiveness (20 Points):** Clear, low-stress UI design, sub-second route calculation, high-contrast disaster visibility, and mobile usability.
4. **Resilience & Fallback Architecture (15 Points):** Robust handling of edge cases, offline map tile caching, and graceful degradation during simulated network blackouts.
5. **Code Standards, Repository & Live Demo (15 Points):** Well-architected GitHub repository, mock disaster scenario scripts/GeoJSON datasets, .env.example, and a working live web demonstration.

6. Submission Guidelines

1. **Code Repository:** Public/Private GitHub repository containing clean branch commits, setup guide, mock emergency GeoJSON datasets, and .env.example.
2. **Live Deployment:** Hosted interactive web application (Vercel / Render / Netlify / AWS).
3. **Demo Video / Deck:** A crisp 3-minute video showing:
 4. Triggering an SOS from a citizen's coordinates.
 5. The Commander's console drawing an active hazard geofence (e.g., flash flood zone).

6. The navigation engine dynamically recalculating a safe escape corridor to the nearest open relief shelter.
7. Real-time geofence proximity alert triggering when approaching a closed road.